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Artificial Intelligence: The Future of Mankind

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Artificial Intelligence: The Future of Mankind

Artificial Intelligence, more frequently referred to as artificial intelligence, is a subject that has amassed a lot of attention in recent years. Artificial Intelligence as a field of study has since grown tremendously and has been ranked as one of the fastest-growing fields. Although it is in its years of infancy, artificial intelligence is already generating a trillion dollars a year in revenue (Russell and Norvig, 2021). Though it has seen remarkable growth and is frequently discussed and debated, very few people clearly and concisely understand what artificial intelligence truly is. The reason several people fail to truly grasp the definition of artificial intelligence is that it spans a wide variety and complexity of subfields. In the simplest definition, artificial intelligence seeks to make computers capable of thinking and reasoning just as a human being would. To achieve this difficult feat a computer would have to be able to communicate successfully in a human language, store what it knows or hears, answer questions, draw new conclusions, adapt to new circumstances, and detect and understand logical patterns (Boden, 2018). Artificial Intelligence is by no means a new idea; scientists and researchers alike have theorized and debated the notion of a fully intelligent machine for years. Alan Turing first proposed this idea in his paper "*Computing Machinery and Intelligence*" in which he famously said, "Can machines think?" In this paper, Turing also conceived what is known today as the Turing Test an assessment of a machine's ability to think intelligently in the same regard that a human being would. (Turing, 1950) Ever since Turing's paper in 1950, artificial intelligence has gradually progressed and transitioned from a theoretical concept into a reality. Artificial Intelligence finally gained its name and widespread formal recognition in 1956 at the Dartmouth Workshop in what is regarded today as the "birth of A.I." (Jørgen Veisdal, 2019). Artificial intelligence stands to impact every aspect of human life today. It. has the potential to revolutionize the way we solve problems and our very perception of intelligence as a whole. However, it brings along with it significant risks and dangers in how we live our lives (Livingstone and Risse, 2019). In this essay, we shall take a deep dive into the many facets of artificial intelligence including machine learning, ethics, and morals. We shall discuss the impact of artificial intelligence on the work economy, healthcare services, society, and the future of the human race.

Artificial Intelligence is oftentimes confused with machine learning. Machine learning is the ability of a computer or program to improve its behavior through diligent study of past experiences and to be able to make accurate predictions about the future (Bostrom, 2017). Machine learning can best be understood by the example of a self-driving car software that learns by watching a human driver. As the software watches the driver make more decisions, it begins to form connections and patterns from the data it receives. It slowly understands which side of the road to drive on when to brake, how traffic lights work, and soon enough it has a full grasp of how to drive merely by observation and learning from pre-existing conditions. The significance that machine learning in artificial intelligence holds is that it gives a computer or machine the unique ability to adapt to unforeseen situations. Computers today are typically programmed to complete a specific task and are only capable of doing it in a certain manner (Stuart Jonathan Russell, Norvig and Al, 2010). Machine learning, on the other hand, gives a computer the power to solve problems and solutions that it is not familiar with. This uniqueness and adaptability in artificial intelligence breaks past the traditional limits of human learning and problem-solving. Computers can continuously learn and improve without the need for breaks, creating exponential transformation and improvement in a much shorter time. Computers in essence, also have the ability to commit every piece of information and data to memory-making it far superior in solving problems. The possibilities that machine learning creates in solving complex tasks and developing solutions to problems far surpass the abilities of any known man (Bostrom, 2017). Machine learning extends beyond the bounds of problem-solving and grants a computer incomprehensible power. A computer that has the ability to learn, is also, in turn, able to rewrite its own software. Intelligent machines can detect their weaknesses and design flaws and redesign and reengineer themselves to become more accurate, efficient, all overall more intelligent (Boden, 2018). This advanced and self-improvement would normally require decades of research and development carried out by human engineers. However, all this can be done in mere seconds and moments, and all without the manual input of a human being. This unparalleled power and intelligence brought by machine learning do not merely end there; machine learning allows computers to imitate and successfully replicate some of the most abstract human qualities such as creativity. Many discredit the idea or notion that artificial intelligence can ever truly possess a creative nature, however many examples of artificial intelligence engines today have shown a great affinity and ability to create unique and inexplicably creative works of art. The intelligence engine GPT-3 is able to create realistic text that is almost indistinguishable from human creative writings and it does so with only a short sentence prompt. GPT-3 is capable of writing deeply compelling and emotional stories that resonate with more nuanced human emotions. It is able to achieve this by repetitiously reading and scanning through vast libraries of English literature and forming neural -connections and patterns between ideas, thoughts, words, and sentences[1]. GPT-3 is merely one example of the vast applications of artificial intelligence and machine learning in creative pursuits. Other more developmental artificial engines are able to create realistic CG (computer-generated) art from simple one- or two-word prompts. Even with tremendous advancements in the development of creative artificial intelligence, there exist limitations in how far a machine can go creatively. Creativity in itself can be regarded as the highest form of intelligence in human beings, and yet we still find it difficult to concretely define creativity; much less to replicate it precisely. Creativity in humans lies in the deepest subconscious regions of our brain and is hard to pinpoint precisely. For this reason, creativity in machines is limited in how far they may truly be able to go (Boden, 2018). The irony here is that creativity in artificial intelligence seems closer than ever and yet when compared to the inimitability of human creativity it seems as far away as possible. One of the most contested and debated questions regarding artificial intelligence is whether a machine regardless of its

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[1] New York Times (2022). Meet GPT-3. It Has Learned to Code (and Blog and Argue). (Published 2020). *The New York Times*. [online] Available at: https://www.nytimes.com/2020/11/24/science/artificial-intelligence-ai-gpt3.html [Accessed 23 Mar. 2022].

intelligence can ever learn to exhibit true and genuine emotion, an immutable concept in interacting and dealing with human beings. Emotion, much like creativity is too complex to narrow down to one region of our brain or any neural synapse. Emotion is abstract, but a crucial ability for any intelligent life form. An intelligent machine without true emotion can pose significant safety as well as moral dangers when ethical questions come to play. For example, how does artificial intelligence weigh the value of human lives without emotion, and how does it solve ethical dilemmas? A machine that is incapable of true reason would rely purely on ration and logic to answer these questions, but would this be the best approach? (Mohaghegh, 2021) Machine learning is a vital and crucial factor of artificial intelligence. Machine learning defines how artificial intelligence interprets data, makes complex decisions, and understands more nuanced expressions of intelligence such as creativity and human emotions.

With the rapid advancement of artificial intelligence and its power to radically transform every aspect of human lives, the question of how the technologies will impact businesses, consumers, and the economy as a whole is inescapable. Artificial Intelligence is expected to have a significant influence on global consumer, business, and government sectors (Boden, 2018). According to the McKinsey Global Institute, by 2030, more than 70% of firms will have adopted at least one form of artificial intelligence in their day-to-day operations. According to McKinsey, artificial intelligence may create an additional $13 trillion in economic output by 2030, increasing global GDP by around 1.2 percent each year. This will most likely result in a major disruption on manual labor workers. According to the majority of research, the economic influence and impact of artificial intelligence will be largely positive (Chen et al., 2016). Artificial intelligence coupled together with robotics will revolutionize manufacturing; making production line processes more efficient and productive. Mundane and repetitive tasks like weekly report-writing and accounting responsibilities will all be automated and computerized with astronomically high speed and productivity gains. This automation will affect a large number of jobs; virtually leaving millions of workers and employees without jobs. Artificial Intelligence will render thousands of jobs that may have otherwise even required higher education degrees all but useless. This poses a threat for several people, but along with the dissolution of several jobs, artificial intelligence will birth several new jobs. Artificial intelligence will push the work economy from less manual and technical jobs to more inspired and imaginative jobs (University of St. Thomas, 2020). Innovation and technology are certainly changing. Today, the skills and professions we know must change fundamentally. Our way of doing things and how we perceive jobs is being challenged in a way we have not experienced before; guidelines and rules are changing, causing discomfort, uncertainty, and anxiety. How can we lead the way if we are not sure that traditional methods of hard work, and educational qualifications, do not necessarily guarantee a certain quality of life? According to various reports, warnings indicate that artificial intelligence could lead to the loss of tens of millions of jobs. The question is, when or in what period of time will the acceptance of artificial intelligence and job loss become a reality? We must fundamentally change the way we perceive the job economy today and grow to adapt alongside artificial intelligence (OpenMind, 2020). Automation and technology have changed work to reduce costs, increase efficiency and increase production for several years now. Car transport by horses and carriages, electricity or fluorescent lights, in many cases, have replaced gas lamps and gas has replaced coal. In the past, jobs have shifted and changed with the rapid expansion and development of new technologies and artificial intelligence is no exception. Artificial intelligence calls for the adaptation of jobs as we know them today to accommodate the fast-growing possibilities of artificial intelligence.

Emerging technologies in the form of artificial intelligence and machine learning have begun to change educational tools and institutions. Artificial intelligence possesses the tremendous power to change the future of education. Artificial intelligence in education in the United States is expected to grow by 47.5% between 2017 and 2021, according to the US Education Artificial Intelligence Market Report. Although most experts believe that the critical presence of teachers is irreplaceable, there will be many changes in the work of the teacher and the best teaching methods. Artificial intelligence is not the only way teachers can do their jobs. It also transforms the way students learn. This evolution is only not limited to the United States. According to market research motor, the annual growth rate of Amnesty

International in education increased by 45 percent and is expected to reach 5.8 billion dollars by 2025. Artificial intelligence has already been implemented in the education sector in some institutions by helping to develop skills and test systems. Since artificial intelligence Academy solutions are adults, hope that Amnesty International Plug periods can help with learning and teaching and allows schools and teachers to do more than ever before. artificial intelligence can pay efficiency functions, personal sign, and manager so that teachers can get permission to understand and adapt to time and freedom - unique human possibilities where the machine can struggle. It is given to modify learning based on specific needs of individual students for teachers for many years, but artificial intelligence is an unstable teacher of teachers who need to manage many students in each chapter. Synthetic intelligence tools are useful for including all global lessons, including all global lessons, visual acuities, or hearing damage, including those implementing other languages. Presentation translators are the free additional components of PowerPoint, which generate translations to actual time for the teacher to say. It also opens the likelihood of students that may not be able to go to schools in this disease or cannot learn from a particular subject that cannot be used at school. Educators send tremendous homework and tests. artificial intelligence proposes a recommendation for how to close the gap during learning and performs immediate work for these tasks. The machine can already be a multiple selection test, but it is very close to the written response to evaluate. When artificial intelligence steps automate administrative tasks, the teacher must spend more time with every student. Artificial intelligence often creates more efficient registration and admission procedures. Artificial intelligence is changing transport as well. By allowing cars, trains, ships, and planes to operate autonomously, making them convenient for traffic, they have been used in a variety of transportation applications. As well as making our lives easier, it can also help make all modes of transport safer, cleaner, smarter, and more efficient. Autonomous transport driven by intelligent vehicles, for example, can help reduce human errors associated with various vehicle accidents. However, these opportunities present real problems, including misuse and adverse consequences such as cyberattacks and transport decisions. There are also ethical concerns about the impact of work and responsibility on decisions made by intellectuals on behalf of human beings.

A sector in which artificial intelligence can have the most positive impact on human lives is healthcare. There is already a lot of research that suggests that artificial intelligence could be better or as good as humans in important medical areas such as disease diagnosis. Today, algorithms can mainly be used by radiologists to help researchers diagnose cancer and generate cohorts for costly clinical trials. However, for a number of reasons, it is most likely that it will be years before artificial intelligence replaces humans in a variety of areas of the medical process. We shall discuss in this paragraph, the potential of artificial intelligence to automate aspects of healthcare and some of the hurdles standing in the way of rapid artificial intelligence implementation in healthcare. IBM's artificial intelligence engine named Watson has recently received a lot of media attention with its focus on appropriate drugs, particularly cancer diagnosis and treatment. Watson uses a combination of machine learning and natural language processing technology. However, initial enthusiasm for applying this technology faded away when customers realized how difficult it was to teach Watson how to fight certain types of cancer and apply Watson into treatment processes and systems. care. The problem of implementing artificial intelligence eliminates many health organizations. The problem of integration can be a wider obstacle to prevailing and space executions than can provide accurate and effective recommendations. Many features are based on artificial intelligence to diagnose and treat a technical company that responds to only one simple case. Furthermore, the use of artificial intelligence in medicine has several ethical implications. Health decisions have historically been made primarily by individuals, and the use of intelligent machines to help or assist them raises accountability, transparency, satisfaction, and privacy issues. Perhaps the most difficult problem is addressing the transparency of current technology. If the patient notices that the image leads to a cancer diagnosis, they'll probably want to know why. Even doctors familiar with deep learning algorithms and their practice, in general, may not be able to provide explanations. With artificial intelligence systems, errors in diagnosing and treating a patient are inevitable, and identifying their responsibilities can be difficult. Patients can also receive medical information from an artificial intelligence system that they wish to receive from an empathetic customer. Machine learning systems in healthcare can also be affected by algorithm bias. Algorithm bias can predict an increased likelihood of disease based on gender or race when these are not the true causative factors. Many changes can be observed in artificial intelligence technologies in the fields of behavior, medicine, work, and health. It is important that healthcare institutions, such as governments and regulators, establish structures and act responsibly to manage critical issues and introduce governance mechanisms to reduce these negative impacts. As one of the most potent and potent technologies to affect human society, it will require constant attention and thoughtful policy for years to come. The biggest challenge for artificial intelligence in these areas of health is not whether the technology has enough capability to be useful, but rather ensuring its adoption in routine clinical practice These challenges will eventually be addressed, but it will take more time as the technologies mature. As a result, it is likely to see the use of artificial intelligence in clinical practice over the next five years and greater use over the next ten years. It is also becoming increasingly clear that artificial intelligence systems will not largely replace human clinics, but will increase the effort involved in patient care. Over time, humanities clinics can work on tasks and work schedules that reflect unique human skills such as empathy, persuasion, and integration with the bigger picture. Perhaps the only healthcare providers who will lose their jobs over time are those who refuse to work with artificial intelligence.

In recent years, the ethics regarding artificial intelligence have become an essential issue that those at the very forefront of the advancement of artificial intelligence have begun to ponder and debate. Artificial intelligence is an enormously powerful tool that is set to transform the world as we know it today, and equally with that power for positive change comes the power for great negative change as well. Artificial intelligence if not misused has the ability to destroy the lives of billions and erode the years of positive human civilization and development in all but a few decades. That is why an ethical, as well as a moral approach and understanding in dealing with artificial intelligence, is crucial. In recent years, the introduction of face recognition, analysis, and analysis technology has increased significantly. However, in practice, studies have repeatedly shown that these systems more accurately approximate some populations than others. The problem is that facial recognition and analysis systems are constantly being trained on distorted or biased data. The images of dark-skinned men and women are much smaller than the images of dark-skinned men and women. Even the fairest and most appropriate systems can be used to violate the civil liberties of people. “Without algorithm justice, the technical accuracy/integrity of an algorithm can turn an artificial intelligence tool into a weapon,” says Bulamovini. Predictive models developed with artificial intelligence and machine learning algorithms are based on learning from data. Since we know how this data is used to create artificial intelligence-based models, the primary ethical goal of artificial intelligence is to study how artificial intelligence models change based on the quality and quantity of the data used (Researcher SD, 2021). The quality and quantity of the data used to build an artificial intelligence-based model determine whether the model is biased. The purpose of artificial intelligence ethics is to identify the quality and quantity of the data used to build a model and whether the model data has biases, both intentional as well as unintentional. Bias can enter a deep educational process at many stages, and basic computer science practices are not designed to publish. Or present It reflects distortion. The first case can occur, for example, when a deep learning algorithm delivers more photos of lighter faces than darker ones. The resulting facial recognition system is inevitably worse for darker skin type facial recognition. The second case happened when Amazon discovered that its internal recruitment tool consisted of candidate removal. It learned how to do it because it was trained in historic recruitment decisions that prioritize men over women. When we use the word "ethical" when talking about artificial intelligence we are referring to accepting artificial intelligence transparently, responsibly, and responsibly. For others, this means ensuring that the use of artificial intelligence is adhered to by laws, regulations, ethics, customer expectations, and organizational values. Ethical artificial intelligence also prevents the use of fraudulent data or algorithms and ensures that automated decisions are honest and transparent. Since the invention of technology, it has undoubtedly influenced human behavior, morals, and identity. Technology has sparked a widespread social movement and changed people's perceptions of how they view life. But a question about artificial intelligence arises. Can artificial intelligence threaten human ethics and morals? Technological progress is rapid and people without technology cannot survive in modern society. With the widespread use of technology, the link between technology and ethics becomes more and more inseparable. The idea of a machine that is beyond human can be related to a conscious machine. Excellence in humans means imitating, feeling, and distinguishing different qualities that are important to humans, such as a high recognition value in terms of perceived vision. But can we compare computers to people? Can computers be intelligent? Do computers exceed human capabilities? These are contradictory and controversial questions, especially when there are so many hidden assumptions and misconceptions about how the brain perceives things. The idea of realizing and pursuing human abilities includes knowledge of the various processes and characteristics that define humanity such as intelligence, language, abstract thinking, art, music, emotions, physical abilities, etc. (Signorelli CM, 2018).

A frequent debate between artificial intelligence enthusiasts and experts alike is in regards to the extent of “intelligence” in machines. Several people question a computer’s ability to truly possess intelligence, regardless of its cognitive abilities or ability to respond to new situations. Others on the other hand vehemently defend artificial intelligence; stating that it is in fact “intelligent.” This debate has sparked the creation of a new term known as AGI or Artificial General Intelligence. Artificial general intelligence refers to the ability of an intelligent machine to understand and learn any piece of information that a human being can and to exercise this intelligence freely and without restrictions as a human being would do as well. This definition birthed another definition called Narrow AI, also known as weak AI, where a machine although intelligent is only capable of accomplishing a singular, specific task. On the other hand, artificial general intelligence also referred to as Strong AI, is in all facets indistinguishable from human intelligence and can perform everything that a human being is able to perform with no limitation at all. The contrast of Narrow AI and Artificial General Intelligence has caused many to ask what if a machine is capable of truly possessing human intelligence to the fullest. Some have further gone on to debate what intelligence in beings is and if a machine is really capable of possessing intelligence or just merely imitating the appearance of intelligence. This question, “Can a machine truly be as intelligent as human beings?” is an important one that we must ask ourselves. Many people would be quick to answer this question with an outright “No,” but the answer to such a question is far more complicated than that. This question is more of a philosophical question than it is a scientific or even a technological answer. To answer this question we go back to Alan Turing’s infamous Turing Test of 1950. The Turing Test is simple in nature; it asks whether someone can distinguish, 30 percent of the time, whether they are interacting (for up to five minutes) with a computer or a person. If the person was not able to, Turing believed, then that was empirical evidence that a computer could indeed think and possess intelligence. The underlying fault with Turing’s belief is that it implies that the only qualification of a machine to be intelligent is merely its ability to successfully mimic human behavior and intelligence. Although at the time Turing’s test was revolutionary, today it no longer holds up to the urgent philosophical questions on what really makes a machine intelligent. Several years later in 1980, American philosopher John Searle created a new thought experiment, aimed at answering this same question, in his paper “Minds, Brains, and Programs” which was published in the *Behavioral and Brain Sciences* journal. John Searle’s thought experiment named the Chinese Room Argument goes as follows; “Suppose that artificial intelligence research has succeeded in constructing a computer that behaves as if it understands Chinese. It takes Chinese characters as input and, by following the instructions of a computer program, produces other Chinese characters, which it presents as output. Suppose, says Searle, that this computer performs its task so convincingly that it comfortably passes the Turing test: it convinces a human Chinese speaker that the program is itself a live Chinese speaker. To all of the questions that the person asks, it makes appropriate responses, such that any Chinese speaker would be convinced that they are talking to another Chinese-speaking human being.” The question now that Searle poses to us is “Does this machine understand the Chinese language?” The obvious answer is no. Searle, through the Chinese Room, thought experiment, came to the conclusion that artificial intelligence, therefore, does not truly possess intelligence. Searle argues that for a machine, or any being for that matter to be intelligent, it must be able to demonstrate understanding and intentionality. As Searle famously put it, “The reason that no computer program can ever be a mind is simply that a computer program is only syntactical, and minds are more than syntactical. Minds are semantical, in the sense that they have more than a formal structure, they have a content.” As with all debates, several arguments and counter-arguments have been made supporting or denying the ability of a machine to possess intelligence. No one true answer exists to this daunting question, but this in essence is the philosophical debate between whether a machine can truly be referred to as intelligent.

Having discovered the vast and far-reaching effects of artificial intelligence, the natural question that arises is “What will the next millennia affected by artificial intelligence look like?” As one would assume, there is no one answer to this question. Several experts have pondered this question and have arrived at different beliefs. We will look over some of the different belief models of how our future will look like, shaped by artificial intelligence. The most common beliefs fall under the models of Digital Utopianism, Techno-skepticism, and the Beneficial AI Movement (Tegmark, 2017). Digital utopianism arises from the conviction that technology – defined as more than just tools and machinery – may be used to create a 'perfect' society in the near future. Furthermore, digital utopians usually believe that technology can cure increasingly troubling psychological problems such as violence, rudeness, and social disorder. Digital utopianism creates an envisioned society in which human beings and artificial intelligence live side-by-side in perfect harmony in a world free of all strife, disparity, or even suffering. This digital utopia is achieved by the positive effect of artificial intelligence and technology as a whole (Tegmark, 2017). Techno-skepticism on the other hand is the complete and total opposite of digital utopianism. Techno-skeptics believe that artificial intelligence and more broadly artificial intelligence will be the utter destruction of mankind and human life and prosperity. They often refer to “the singularity” an event in which artificial intelligence will surpass human intelligence and consequently human capability. Technology will take over every facet of life and shall render human beings useless. In this bleak and gloomy future, artificial intelligence rebels against man, ultimately ending human civilization. Techno-skeptics preach that artificial intelligence should not be attempted to be created and that in doing so the path towards our own destruction is inevitable (Tegmark, 2017). The beneficial AI movement stands in the moderate balance between digital utopianism and techno-skepticism. The beneficial AI movement, much like digital utopianism, is the belief that artificial can make the world a significantly better place and can meaningfully impact human civilization. This belief is followed with the caveat that severe regulations are taken in the development of artificial intelligence and that we must be careful in how far we allow artificial intelligence to go. Without these regulations, then the world can also be severely damaged and negatively impacted all the same. The beneficial AI movement advises us to be cautious of the immense power that artificial intelligence carries and that in the wrong hands it can do much harm and yet in the right hands it can do such tremendous good (Tegmark, 2017).

We have seen just how powerful artificial intelligence can be. The question now is not whether artificial intelligence can happen or not. The real question is "Will artificial intelligence create a positive or negative impact in the world?" The timeline for how artificial intelligence develops is not certain; we are not sure how many years it may take for artificial intelligence to equal human intelligence. But we are sure of the monumental effect that artificial intelligence in tandem with developing technology will have. In line with the belief of the beneficial AI movement, we must be vigilant even with artificial intelligence still within its relative years of infancy. Artificial intelligence, like all other technological advancements before it, must be viewed as a tool wielded by humans for our progression and development. Artificial intelligence should not be feared or unwisely deemed as the "end of times" and destruction of the human race. At the advent of every major technological development in history were always nay-sayers who were skeptical of how such advancements would change the world. Change, even that brought about by artificial intelligence is inevitable and we must embrace it rather than run away from it. Artificial intelligence, as we have seen throughout this essay.

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